

Business Case: Target SQL

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Overview: Target is a globally renowned brand and a prominent retailer in the United States. Target makes itself a preferred shopping destination by offering outstanding value, inspiration, innovation and an exceptional guest experience that no other retailer can deliver.

This particular business case focuses on the operations of Target in Brazil and provides insightful information about 100,000 orders placed between 2016 and 2018.

As a data analyst I have analyze the following data. I have provided few insights and recommendations based on my analysis

1 --- Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

1. Data type of all columns in the "customers" table.

SCHEMA

DETAILS

PREVIEW

TABLE EXPLORER

PREVIEW

INSIGHTS

LINEAGE

DATA PROFILE

DATA QUALITY

Filter

Enter property name or value

☐	Field name	Type	Mode	Key	Collation	Default Value	Policy Tags ?	Description
☐	customer_id	STRING	NULLABLE	-	-	-	-	-
☐	customer_unique_id	STRING	NULLABLE	-	-	-	-	-
☐	customer_zip_code_prefix	INTEGER	NULLABLE	-	-	-	-	-
☐	customer_city	STRING	NULLABLE	-	-	-	-	-
☐	customer_state	STRING	NULLABLE	-	-	-	-	-

EDIT SCHEMA

VIEW ROW ACCESS POLICIES

Insight: There are 5 columns in the 'customers' table. Where 4 of them has string type value and 1 has integer type value

Recommendation: NA

2. Get the time range between which the orders were placed.

```
select min(order_purchase_timestamp) as min_timestamp,  
       max(order_purchase_timestamp) as mx_timestamp  
from `ABC.orders`
```

Row	min_timestamp ▼	mx_timestamp ▼
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC

Insights: The very first order was placed on 4th of Sept of 2016 at 9:15 PM in the UTC time zone. And the last order was placed on 17th of Oct of 2018 at 5:30 PM in the UTC time zone.

Recommendation: NA

3. Count the Cities & States of customers who ordered during the given period.

```
select count(distinct customer_city) as city,  
       count(distinct customer_state) as state  
from `ABC.customers`  
where customer_id in (select customer_id  
                      from `ABC.orders`)
```

city ▼	state ▼
4119	27

Insights: As we can see within Sept 2016 and Oct 2018 we have got orders from 27 states and 4119 cities in total. It means we were able to spread and run the business in all over Brazil.

Recommendation: NA

2 --- In-depth Exploration:

1. Is there a growing trend in the no. of orders placed over the past years?
2. Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

```
select extract(year from order_purchase_timestamp ) as year,
       extract( month from order_purchase_timestamp ) as month,
       count(order_id) as orders
from `ABC.orders`
group by 1,2
order by 1,2
```

Row	year	month	orders
1	2016	9	4
2	2016	10	324
3	2016	12	1
4	2017	1	800
5	2017	2	1780
6	2017	3	2682
7	2017	4	2404
8	2017	5	3700
9	2017	6	3245
10	2017	7	4026

Insights: We have started getting orders from the month of sept of 2016. However, in this starting phase of the business we can see a significant amount of orders in the month of oct in 2016. It happened due to upcoming holidays in the month of Nov and Dec.

Then from the next year we can see an increasing amount of orders in each month. Again, due to holiday seasons and the popularity of our business we get the highest orders in Nov 2017 and similar to 2016, we got a drop of orders in Dec.

Then in 2018 the orders got stable in each month and no fluctuation noticed. Since, we were closing the business in brazil, the order dropped in Sept and Oct.

Recommendation: In the starting of the business, I believe, we have given a lot of ads and reached out to the buyers through different way – which has resulted the increasing amount of orders in 2017. In 2018, we got orders due to our good will and trust – but we were needed to do much more of marketing. Also we need to focus on Oct and Nov as these are the times when the festive and holiday season starts and buyers tend to place more orders than usual.

2. During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)

- 0-6 hrs : Dawn

- **7-12 hrs : Mornings**
- **13-18 hrs : Afternoon**
- **19-23 hrs : Night**

```
select case when extract(hour from order_purchase_timestamp ) between
0 and 6 then 'Dawn'
        when extract(hour from order_purchase_timestamp ) between 7
and 12 then 'Mornings'
        when extract(hour from order_purchase_timestamp ) between 13
and 18 then 'Afternoon'
        else 'Night' end as time_of_the_day,
count (order_id) as count_of_orders
from `ABC.orders`
group by 1
order by 2 desc
```

time_of_the_day ▼	count_of_orders ▼
Afternoon	38135
Night	28331
Mornings	27733
Dawn	5242

Insights: We have got most of the orders from the second half of the day. where we got most of the order in the afternoon. In the dawn the number of orders is very less – it can happen due to the time being the midnight.

Recommendation: NA

3 --- Evolution of E-commerce orders in the Brazil region:

1. Get the month on month no. of orders placed in each state.

```
select extract (year from order_purchase_timestamp) year_number,
extract (month from order_purchase_timestamp) month_number,
format_date('%B', order_purchase_timestamp ) as month,
customer_state, count(order_id) as count_of_order
```

```

from `ABC.orders` o left join `ABC.customers` c
using(customer_id)
group by 1,2,3,4
order by 4, 1, 2

```

year_number	Mnum	month ▼	customer_state ▼	count_of_order ▼
2017	1	January	AC	2
2017	2	February	AC	3
2017	3	March	AC	2
2017	4	April	AC	5
2017	5	May	AC	8
2017	6	June	AC	4
2017	7	July	AC	5
2017	8	August	AC	4
2017	9	September	AC	5
2017	10	October	AC	6

**Insights and Recommendations are added with next question.

2. How are the customers distributed across all the states?

```

select customer_state,
       count(customer_id) as count_of_customer
from `ABC.customers`
group by 1
order by 2

```

Row	customer_state ▼	count_of_customer
1	RR	46
2	AP	68
3	AC	81
4	AM	148
5	RO	253
6	TO	280
7	SE	350
8	AL	413
9	RN	485
10	PI	495

Insight: We can see that the customers of state- SP has made the most amount of order. Other than that, we have states- RJ, MG, has significant number of orders. And these states have the highest number of customers.

On the other hand, we have state- RR, AC, AP, AM with the least amount of orders. We can also see that the customers are also very less in comparison to other states. Other states have very moderate number of orders.

The states like – PR, SC, BR has high number of customers but the number of orders from these states are very lower than expected.

Recommendation: We need to check and find out, despite having such huge number of customer, why states like PR, SC, BR have such low amount of orders. We need to do more campaigns and surveys to know what barrier they are facing and we need to act accordingly. We also need to make strategy on how to get more customers on states like RR, AC, AP, AM so that we can expect more orders from these states too.

4 --- Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

- 1. Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).**
You can use the "payment_value" column in the payments table to get the cost of orders.

```
with aa as
(select extract (year from order_purchase_timestamp) Year,
      sum(payment_value) as total
 from `ABC.orders` o join `ABC.payments` p
 using(order_id)
 where extract (year from order_purchase_timestamp) in (2017, 2018)
 and extract (month from order_purchase_timestamp) between 1 and 8
 group by 1
 order by 1 )

select year,total,
      round(((lead(total) over (order by year) - total)/total)*100) as
cost_increase
from aa
```

year ▼	total ▼	cost_increase ▼
2017	3669022.119999...	137.0
2018	8694733.839999...	null

Insight: We can see an increase of cost of 137% from year 2017 to 2018 (include months between Jan to Aug only).

Recommendation: NA

2. Calculate the Total & Average value of order price for each state.

```
with abc as
(select customer_state,
        round(sum(price)) as sum_of_price,
        round(count(distinct order_id)) as no_of_orders
from `ABC.orders` o join `ABC.order_items` oi
using(order_id)
join `ABC.customers` c
using (customer_id)
group by 1)

select customer_state, no_of_orders, sum_of_price,
        sum_of_price/no_of_orders as avg_of_price
from abc
order by 2 desc
```

customer_state ▼	no_of_orders ▼	sum_of_price ▼	avg_of_price ▼
SP	41375.0	5202955.0	125.7511782477...
RJ	12762.0	1824093.0	142.9315937940...
MG	11544.0	1585308.0	137.3274428274...
RS	5432.0	750304.0	138.1266568483...
PR	4998.0	683084.0	136.6714685874...
SC	3612.0	520553.0	144.1176633444...
BA	3358.0	511350.0	152.2781417510...
DF	2125.0	302604.0	142.4018823529...
ES	2025.0	275037.0	135.8207407407...
GO	2007.0	294592.0	146.7822620827...

Insight: State SP has the highest number of orders with highest price and state RR has lowest number of orders with lowest price. However, if we check the average we can see that, avg of price for state- SP is the lowest but state- PB has the highest avg of price.

Recommendation: NA

3. Calculate the Total & Average value of order freight for each state.

```
with abc as
(select customer_state,
        sum(freight_value) as sum_of_price,
        count(distinct order_id) as no_of_orders
from `ABC.orders` o join `ABC.order_items` oi
using(order_id)
join `ABC.customers` c
using (customer_id)
group by 1)

select customer_state, no_of_orders, sum_of_price,
        sum_of_price/no_of_orders as avg_of_Price
from abc
order by 2 desc
```

customer_state	no_of_orders	sum_of_price	avg_of_Price
SP	41375	718723.0699999...	17.37095033232...
RJ	12762	305589.3100000...	23.94525231154...
MG	11544	270853.4600000...	23.46270443520...
RS	5432	135522.7400000...	24.94895802650...
PR	4998	117851.6800000...	23.57976790716...
SC	3612	89660.26000000...	24.82288482835...
BA	3358	100156.6799999...	29.82628945801...
DF	2125	50625.49999999...	23.82376470588...
ES	2025	49764.59999999...	24.57511111111...
GO	2007	53114.97999999...	26.46486297957...

Insight: State SP has the highest number of orders but state- SC has highest amount spent. On the other hand state- RR has lowest number of orders with lowest price. But checking the avg we can find that RR has highest average price and SP has lowest.

Recommendation: NA

5 --- Analysis based on sales, freight and delivery time

1. Find the no. of days taken to deliver each order from the order's purchase date as delivery time.

Also, calculate the difference (in days) between the estimated & actual delivery date of an order.

```
select order_id,order_purchase_timestamp,  
order_delivered_customer_date, order_estimated_delivery_date,  
    DATETIME_DIFF(order_delivered_customer_date,order_purchase_timestamp,day)as time_to_deliver,  
    DATETIME_DIFF(order_delivered_customer_date,order_estimated_delivery_date,day)as diff_estimated_delivery  
from `ABC.orders`  
order by order_purchase_timestamp
```

order_id	order_purchase_timestamp	order_delivered_customer_date	order_estimated_delivery_date	time_to_deliver	diff_estimated_delivery
2e7a8482f6fb09756ca50c10d...	2016-09-04 21...	null	2016-10-20 00:00:00 UTC	null	null
e5fa5a7210941f7d56d0208e4...	2016-09-05 00...	null	2016-10-28 00:00:00 UTC	null	null
809a282bbd5dbcab6f2f724fc...	2016-09-13 15...	null	2016-09-30 00:00:00 UTC	null	null
bfb0f9bdef84302105ad712db...	2016-09-15 12...	2016-11-09 07:47:38 UTC	2016-10-04 00:00:00 UTC	54	36
71303d7e93b399f5bcd537d12...	2016-10-02 22...	null	2016-10-25 00:00:00 UTC	null	null
3b697a20d9e427646d925679...	2016-10-03 09...	2016-10-26 14:02:13 UTC	2016-10-27 00:00:00 UTC	23	0
be5bc2f0da14d8071e2d45451...	2016-10-03 16...	2016-10-27 18:19:38 UTC	2016-11-07 00:00:00 UTC	24	-10
65d1e226dfaeb8cdc42f66542...	2016-10-03 21...	2016-11-08 10:58:34 UTC	2016-11-25 00:00:00 UTC	35	-16
a41c8759f7be7aab36ea07e038...	2016-10-03 21...	2016-11-03 10:58:07 UTC	2016-11-29 00:00:00 UTC	30	-25
d207cc272675637bfed0062ed...	2016-10-03 22...	2016-10-31 11:07:42 UTC	2016-11-23 00:00:00 UTC	27	-22

Insight: Null rows indicates that those orders have not been delivered yet. Therefore, we can say a lot of orders from 2016 to 2018 have not been delivered yet. On the other hand, we can also observe that a huge amount of orders has been delivered after the estimated date has passed

Recommendation: We need to focus more on on-time delivery. In order to do that we need to find out the reason why the order is not getting delivered on time. We also need to get more info on the orders that have null delivery values.

2. Find out the top 5 states with the highest & lowest average freight value.

```
with aaa as (  
select customer_state,  
    avg(freight_value) as average,
```

```

        dense_rank() over(order by avg(freight_value) desc) as
highest_rank,
        dense_rank() over(order by avg(freight_value)) as lowest_rank
from `ABC.customers` c left join `ABC.orders` o
using (customer_id)
left join `ABC.order_items` oi
using(order_id)
group by 1
order by 2 desc
)

select *
from aaa
where highest_rank <= 5
or lowest_rank <= 5
order by 2 desc

```

customer_state	average	highest_rank	lowest_rank
RR	42.98442307692...	1	27
PB	42.72380398671...	2	26
RO	41.06971223021...	3	25
AC	40.07336956521...	4	24
PI	39.14797047970...	5	23
DF	21.04135494596...	23	5
RJ	20.96092393168...	24	4
MG	20.63016680630...	25	3
PR	20.53165156794...	26	2
SP	15.14727539041...	27	1

Insight: State- RR,PB,RO,AC,PI has highest average freight_value and state- DF,RJ,MG,PR, SP has lowest average freight_value.

Recommendation: NA

3. Find out the top 5 states with the highest & lowest average delivery time.

Lowest:

```

select customer_state,
        avg(DATETIME_DIFF(order_delivered_customer_date,order_purchase
_timestamp,day))as avg_delivery_time
from `ABC.orders` o join `ABC.customers` c

```

```
using(customer_id)
group by 1
order by 2
```

Row	customer_state	avg_delivery_time
1	SP	8.298061489072...
2	PR	11.52671135486...
3	MG	11.54381329810...
4	DF	12.50913461538...
5	SC	14.47956019171...

Highest:

```
select customer_state,
       avg(DATETIME_DIFF(order_delivered_customer_date,order_purchase
_timestamp,day))as avg_delivery_time
from `ABC.orders` o join `ABC.customers` c
using(customer_id)
group by 1
order by 2 desc
```

Row	customer_state	avg_delivery_time
1	RR	28.97560975609...
2	AP	26.73134328358...
3	AM	25.98620689655...
4	AL	24.04030226700...
5	PA	23.31606765327...

Insight: State- RR, AP, AM, AL, PA has highest average delivery time and state- SP, PR, MG, DF, SC has lowest average delivery time.

Recommendation: We need to check other states to know why average delivery time is lower than the 5 highest states and work on it.

4. Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.

You can use the difference between the averages of actual & estimated delivery date to figure out how fast the delivery was for each state.

```
select customer_state,
       avg(DATETIME_DIFF(order_estimated_delivery_date,order_delivered_customer_date, day)) as diff_day,
from `ABC.orders` join `ABC.customers`
using (customer_id)
group by 1
order by diff_day desc
limit 5
```

Row	customer_state	diff_day
1	AC	19.762500000000...
2	RO	19.13168724279...
3	AP	18.73134328358...
4	AM	18.60689655172...
5	RR	16.41463414634...

Insight: States- AC, RO, AP, AM, RR are really fast in comparison to other states.

Recommendation: NA

6 --- Analysis based on the payments

1. Find the month on month no. of orders placed using different payment types.

```
select extract(year from order_purchase_timestamp) as year,
       extract(month from order_purchase_timestamp) as monthNum,
       format_date('%B', order_purchase_timestamp ) as month,
       payment_type,count(order_id) as OrderCount
from `ABC.payments`left join `ABC.orders`
using(order_id)
group by 1,2,3,4
order by 1,2,5 desc
```

year ▼	monthNum ▼	month ▼	payment_type ▼	OrderCount ▼
2016	9	September	credit_card	3
2016	10	October	credit_card	254
2016	10	October	UPI	63
2016	10	October	voucher	23
2016	10	October	debit_card	2
2016	12	December	credit_card	1
2017	1	January	credit_card	583
2017	1	January	UPI	197
2017	1	January	voucher	61
2017	1	January	debit_card	9

Insight: Most of the orders are made using credit card in each month and lowest is debit card. However, in September and October of 2018 only voucher was used.

Recommendation: NA

2. Find the no. of orders placed on the basis of the payment installments that have been paid.

```
select payment_installments, count(order_id) as total_orders
from `ABC.payments`
where payment_installments >=1
group by 1
order by 2 desc
```

Row	payment_installment	total_orders ▼
1	1	52546
2	2	12413
3	3	10461
4	4	7098
5	10	5328
6	5	5239
7	8	4268
8	6	3920
9	7	1626
10	9	644

Insight: Total 52546 orders have paid at least 1 payment installment. 23 payment installment has been paid for 1 order only.

Recommendation: NA